# Lab 3: Propositional Logic & Model Checking

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## Task 1 - Understand the Code

1.1 harry.py: The knowledge base is true in only one of the eight possible assignments: rain = True, hagrid = False, dumbledore = True.

| rain | hagrid | dumbledore | KB |
|---|---|---|---|
| False | False | False | False |
| False | False | True | False |
| False | True | False | False |
| False | True | True | False |
| True | False | False | False |
| True | False | True | True |
| True | True | False | False |
| True | True | True | False |

Because rain is true in every model satisfying the knowledge base, model_check(knowledge, rain) returns True.

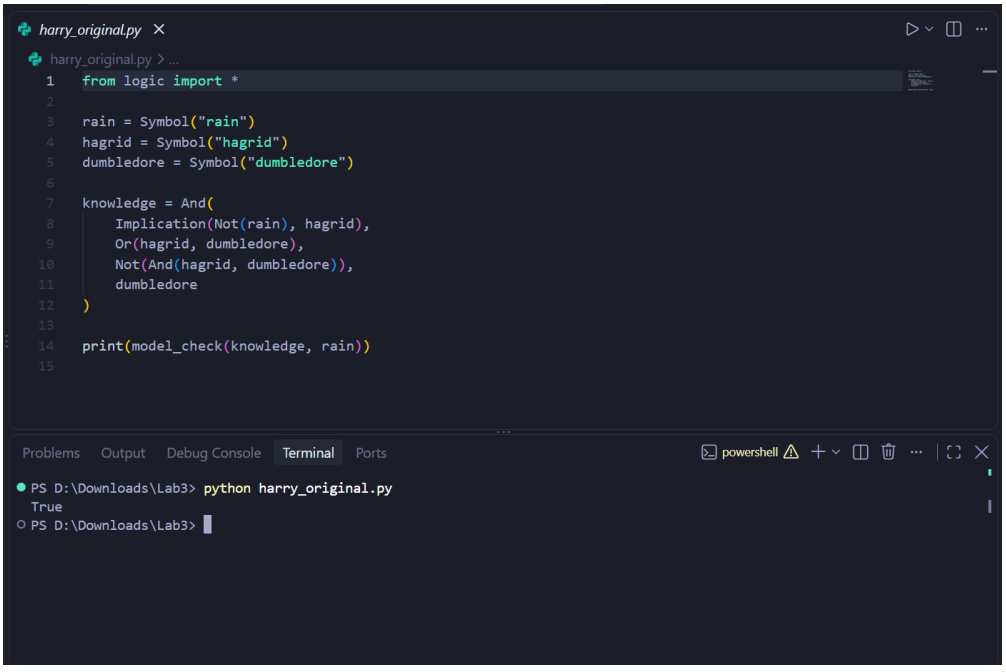



*Figure 1. Original harry.py: True.*

# Task 1 - clue.py

ColMustard and ProfPlum are excluded, so MsScarlet is the suspect. Kitchen and ballroom are excluded, leaving library. Revolver is excluded, and the unknown-card constraint eliminates wrench. The remaining weapon is knife.

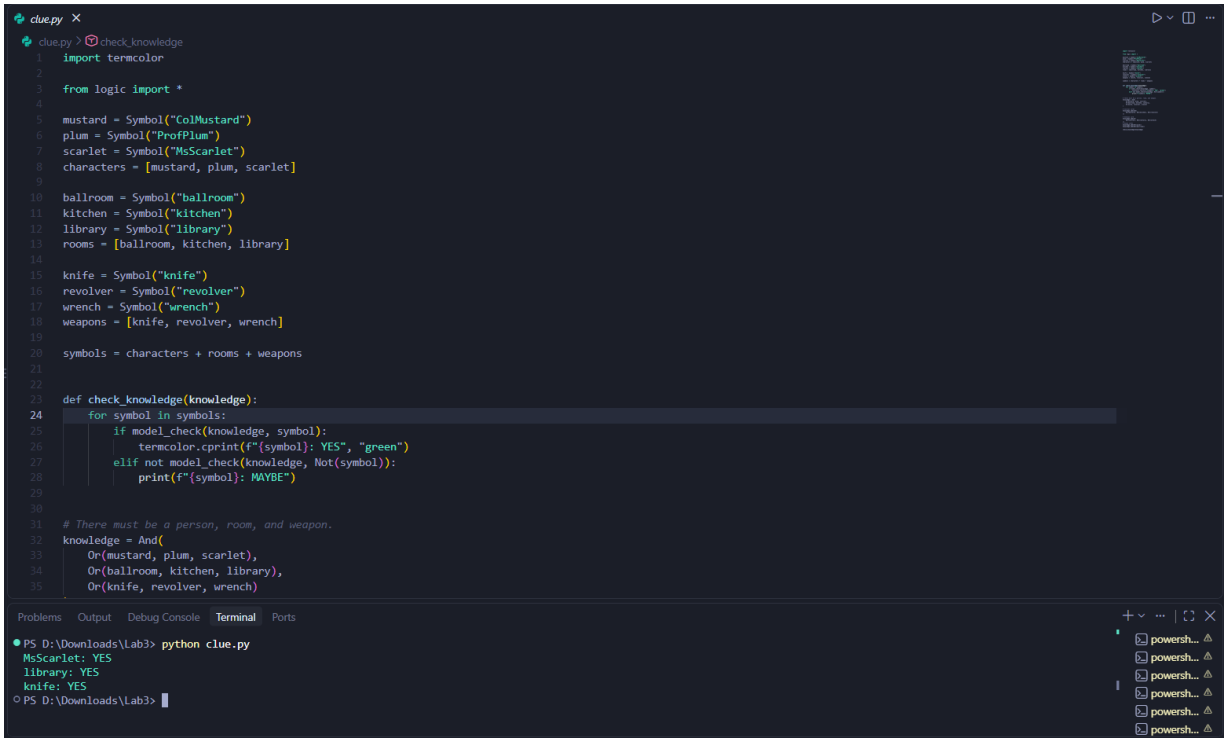



*Figure 2. clue.py: MsScarlet, library, and knife.*

# Task 2 - Modify harry.py

I removed the final dumbledore fact from the knowledge base.

**Prediction: False.  Actual output: False.**

Rain is no longer guaranteed. For example, rain = False, hagrid = True, and dumbledore = False satisfies the remaining rules. Therefore, the knowledge base does not entail rain.

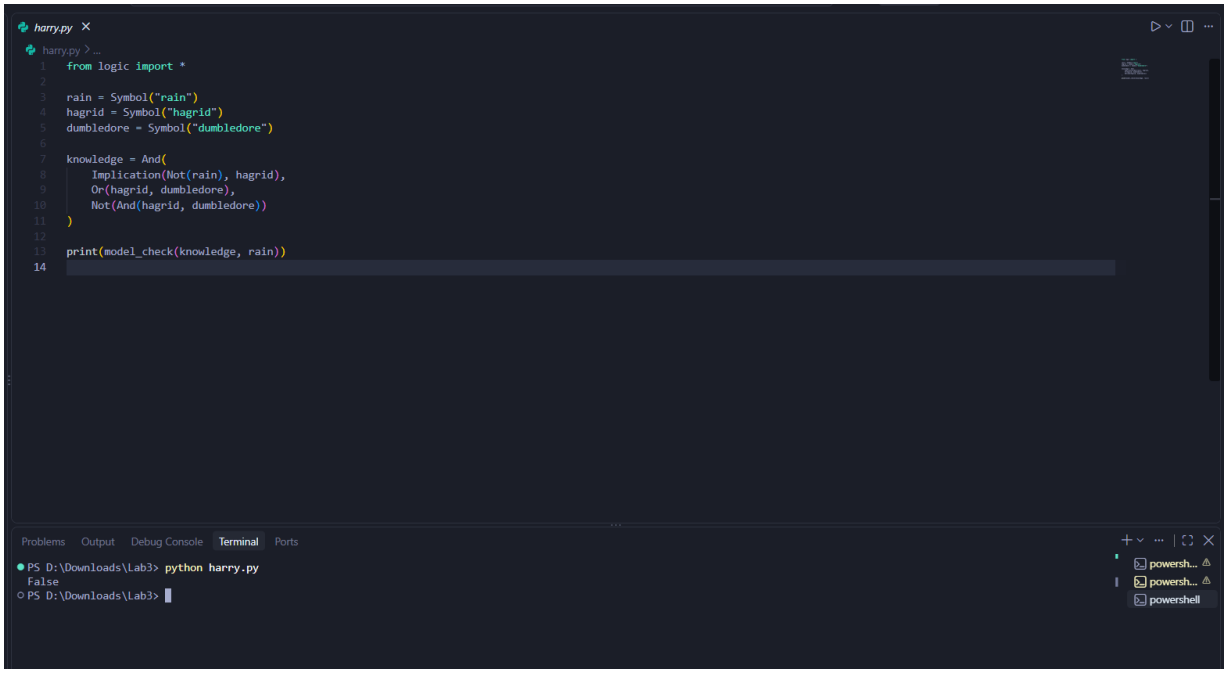



*Figure 3. Modified harry.py: False.*

# Task 2 - Remove the Minerva Clue

I removed MinervaGryffindor from puzzle.py.

**Prediction: No certain house assignments.  Actual output: No assignments printed.**

Eight valid house assignments remain, and no person has the same house in every valid assignment.

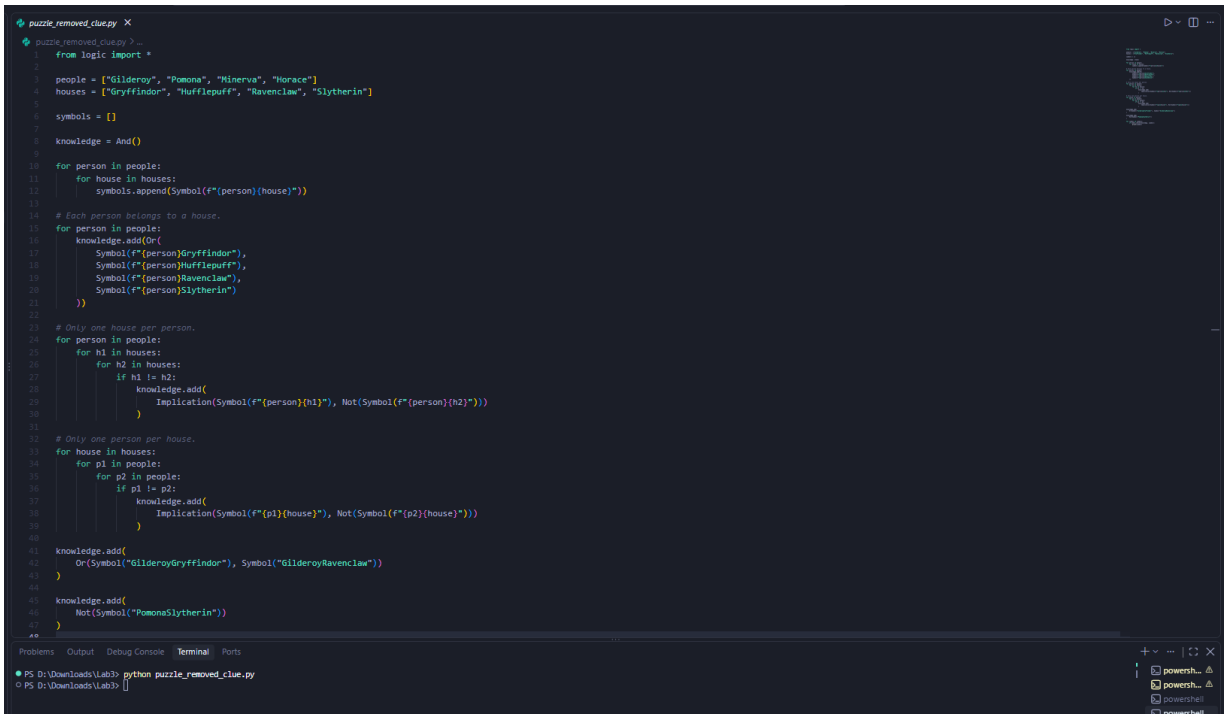



*Figure 4. puzzle.py with the Minerva clue removed.*

# Task 2 - Add a New Clue

I added a compound clue using And: Gilderoy is in Ravenclaw, Pomona is in Hufflepuff, and Horace is in Slytherin. This leaves Gryffindor for Minerva and gives one unique solution.

**Prediction: One complete assignment.  Actual output: All four assignments.**

| Person | House |
|---|---|
| **Gilderoy** | Ravenclaw |
| **Pomona** | Hufflepuff |
| **Minerva** | Gryffindor |
| **Horace** | Slytherin |

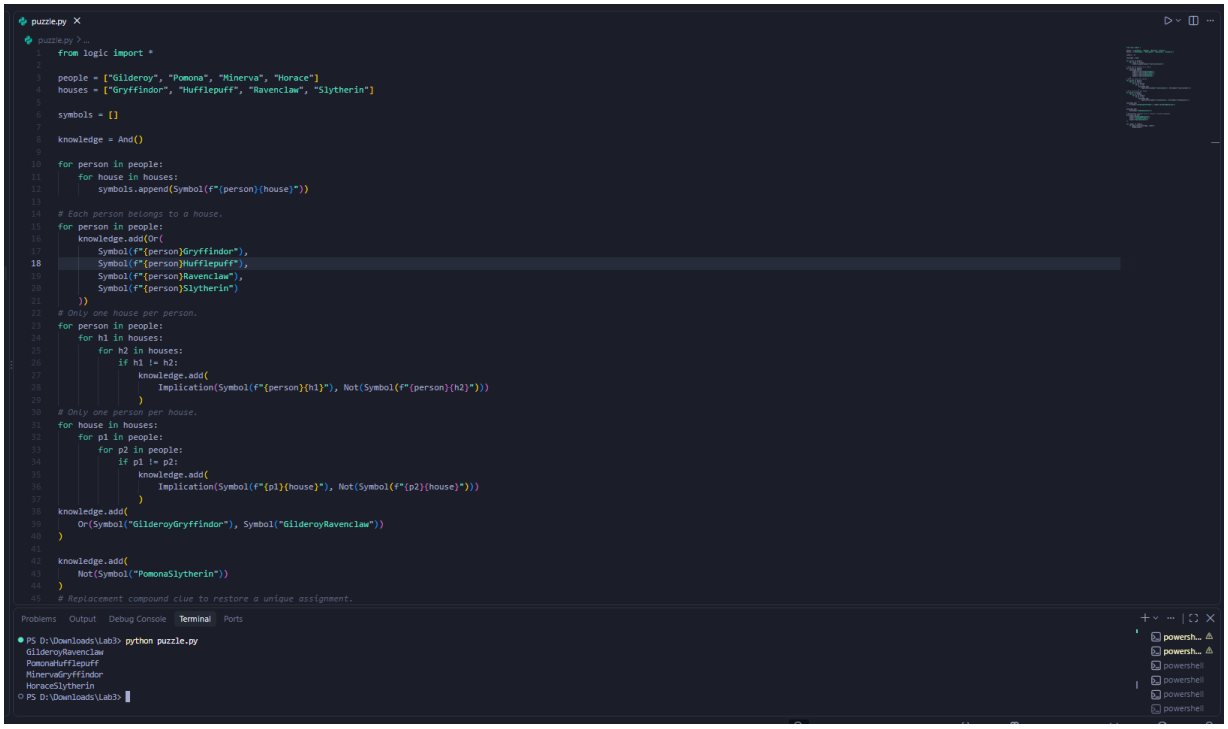



*Figure 5. puzzle.py after adding the compound clue.*

# Task 3 - Knights and Knaves

Each person must be a knight or a knave, but not both. A knight tells the truth, while a knave lies.

| Puzzle | Result |
|---|---|
| **0** | A: Knave |
| **1** | A: Knave; B: Knight |
| **2** | A: Knave; B: Knight |
| **3 (Bonus)** | A: Knight; B: Knave; C: Knight |

Puzzle 0: A cannot be both types, so A lies. Puzzle 1: A cannot truthfully say both are knaves; B is therefore a knight. Puzzle 2: A lies about being the same kind, while B truthfully says they are different.

Bonus Puzzle 3: A said "I am a knight." B lies about what A said, so B is a knave. B also lies about C, making C a knight. C confirms that A is a knight.

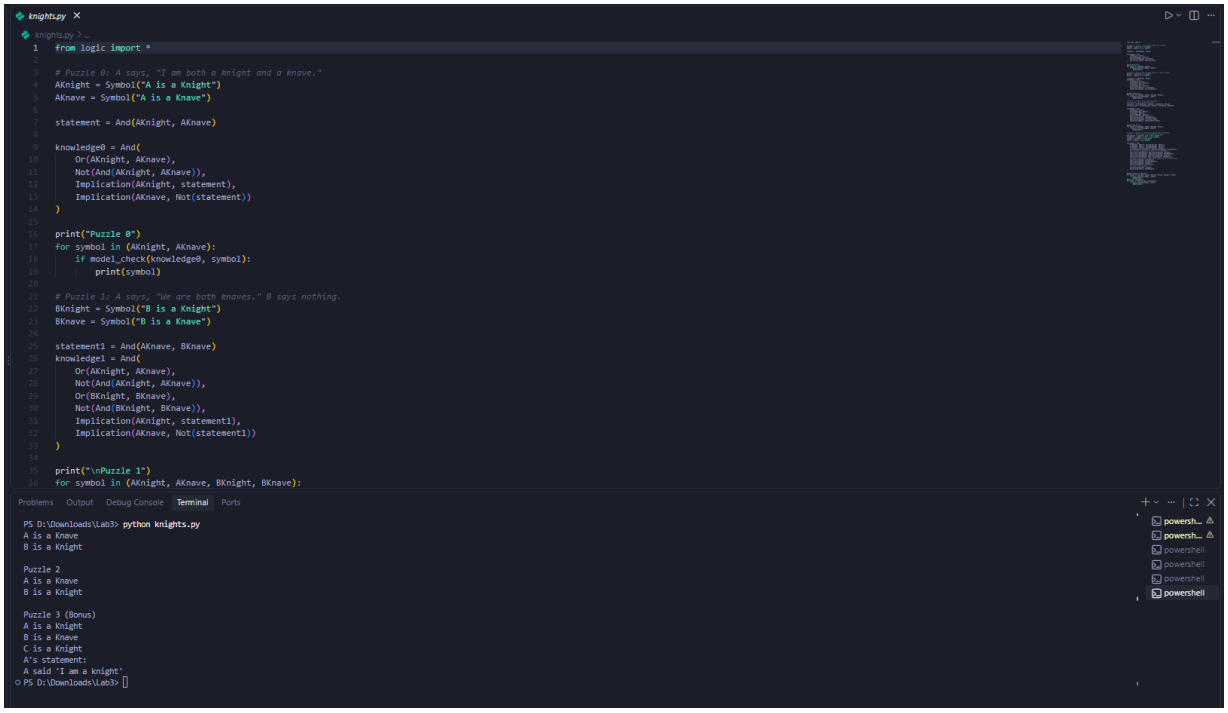



*Figure 6. knights.py results for Puzzles 0-3 (saved output).*

Conclusion: Model checking confirmed the expected results. Changing the knowledge base changed what could be logically proved.